Lecture 9

Functions of random variables

(I) Functions of one random variable.

- Let X be a random variable with pf/pdf $f(x)$.
- Let r be a function defined on the set of possible values of X.
- Let $Y = r(X)$ be a new variable, obtained by transforming X via the function r.
- What is the pf/pdf of Y?

  We denote the pf/pdf of Y by $g(y)$ and calculate it below.

• Finding the p.f. of $Y = h(X)$ if X is discrete

Result 1 If $X \sim f(x)$ and is discrete, then the p.f. of the discrete r.v. $Y = h(X)$, for some given h, is denoted by $g(y)$ and given by:

$$g(y) = \mathbb{P}(Y = y) = \mathbb{P}(h(X) = y)$$

$$= \sum_{\{x:\, h(x) = y\}} f(x).$$

Example $X \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & 9 \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} & \frac{1}{9} & \dots & \frac{1}{9} \end{pmatrix}$.

Let $Y = |X - 5|$, that is $h(x) = |x - 5|$. Then

$$Y = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ \frac{1}{9} & \frac{2}{9} & \frac{2}{9} & \frac{2}{9} & \frac{2}{9} \end{pmatrix}, \text{ as below:}$$

- $\mathbb{P}(Y = 0) = \mathbb{P}(|X - 5| = 0) = \sum_{\{x:\, |x-5| = 0\}} f(x) = f(5) = \frac{1}{9}$
- $\mathbb{P}(Y = 4) = \sum_{\{x:\, |x-5| = 4\}} f(x) = f(9) + f(1) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$

Finding the c.d.f. of $Y = k(X)$ when X is continuous.

Result 2 Let $X \sim f(x)$ be a continuous r.v. Let $Y = k(X)$. The c.d.f. G of Y is:

$$G(y) = \mathbb{P}(Y \le y) = \mathbb{P}(k(X) \le y)$$

$$= \int_{\{x : k(x) \le y\}} f(x)\,dx, \quad \text{for any } y \in \mathbb{R}.$$

Result 3 If Y has a continuous distribution, then the p.d.f $g(y)$ of Y can be calculated as:

$$g(y) = G'(y), \quad \text{at every point } y$$

where G is differentiable.

Theorem 1 (A general result for deriving directly the p.d.f. of $Y = h(X)$ when h is one-to-one and differentiable); Th. 3.8.4 page 172.

- Assume $X \sim$ p.d.f. $f(x)$; $a_1 < X < b_1$, with prob. 1.
- Let $Y = h(X)$.
- Assume the function h is differentiable and one-to-one on (a, b). Denote by $s(y)$ the inverse of $h(x)$; $h^{-1}(y) \equiv s(y)$.
- If h is defined on (a_1, b_1), let (α, β) denote the image of (a_1, b_1) under h.
- Then Y has a p.d.f. g given by:

$$g(y) = \begin{cases} f(s(y)) \cdot |s'(y)|, & \text{for } \alpha < y < \beta \\ 0, & \text{otherwise.} \end{cases}$$

Example Let $X \sim f(x)$, $-\infty < x < \infty$.

Assume: $h(x) = ax + b$, for given $a \neq 0$ and b, for $-\infty < x < \infty$.

- Clearly $(\alpha, \beta) = (-\infty, \infty)$.
- h is differentiable everywhere and invertible.

To find its inverse, solve for x:

$$h(x) = y \iff ax + b = y \iff$$

$$x = \frac{y-b}{a} \implies$$

$$h^{-1}(y) = \frac{y-b}{a} \quad \underline{\text{Notation}} \quad s(y), \text{ for } -\infty < y < \infty.$$

Therefore: $g(y) = \begin{cases} f\left(\frac{y-b}{a}\right) \cdot \left|\frac{1}{a}\right|, & \text{for } -\infty < y < \infty \\ 0, & \text{otherwise.} \end{cases}$

This is summarized below.

Result 4 (The p.d.f. of a linear transformation of a random variable).

Assume $X \sim f(x)$ is continuous.
Let $Y = aX + b$, for given numbers a, b, with $a \neq 0$. Then $Y \sim g(y)$, with

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right).$$

Example 1 Let $X \sim f(x)$:

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y = -2X + 3$. Then $Y \sim g(y)$, with

$$g(y) = \frac{1}{2} f\left(\frac{y-3}{-2}\right) = \frac{1}{2} f\left(\frac{3-y}{2}\right).$$

Example 2 = Example 3.8.4 page 169, next page.

Example 3.8.4 Deriving the p.d.f. of X^2 when X Has a Uniform Distribution. Suppose that X has the uniform distribution on the interval $[-1, 1]$, so

$$f(x) = \begin{cases} 1/2 & \text{for } -1 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We shall determine the p.d.f. of the random variable $Y = X^2$.

Since $Y = X^2$, then Y must belong to the interval $0 \le Y \le 1$. Thus, for each value of Y such that $0 \le y \le 1$, the c.d.f. $G(y)$ of Y is

$$\begin{aligned} G(y) &= \Pr(Y \le y) = \Pr(X^2 \le y) \\ &= \Pr(-y^{1/2} \le X \le y^{1/2}) \\ &= \int_{-y^{1/2}}^{y^{1/2}} f(x)\, dx = y^{1/2}. \end{aligned}$$

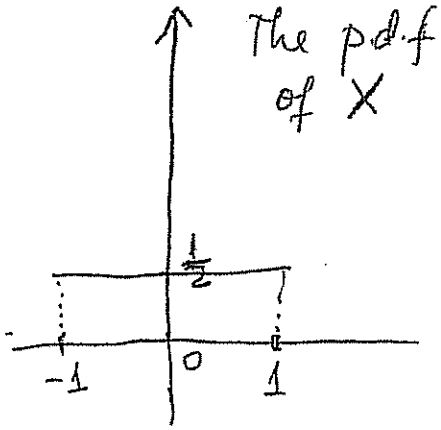
For $0 < y < 1$, it follows that the p.d.f. $g(y)$ of Y is

$$g(y) = \frac{dG(y)}{dy} = \frac{1}{2y^{1/2}}.$$

This p.d.f. of Y is sketched in Fig. 3.23. It should be noted that although Y is simply the square of a random variable with a uniform distribution, the p.d.f. of Y is unbounded in the neighborhood of $y = 0$. ◀

Figure 3.23 The p.d.f. of $Y = X^2$ in Example 3.8.4.





Note : We cannot use Theorem 1 for this calculation because $x(x) = x^2$ is not invertible, for $x \in [-1, 1]$.

How to generate values of a random variable with a given c.d.f. F ?

Theorem

(a) Assume that the c.d.f. of a continuous r.v. X is F. Construct $Y = F(X)$. Then: $Y \sim \text{Unif}[0,1]$.

(b) Let $Y \sim \text{Unif}[0,1]$.
Let F be any continuous c.d.f. with quantile function F^{-1}. Then:

(*) $X = F^{-1}(Y)$ has c.d.f. F.

Jargon: The transformation in (a) is called the probability integral transformation. An illustration is Ex 3.8.5.

The Probability Integral Transformation

Example 3.8.5 Let X be a continuous random variable with p.d.f. $f(x) = \exp(-x)$ for $x > 0$ and 0 otherwise. The c.d.f. of X is $F(x) = 1 - \exp(-x)$ for $x > 0$ and 0 otherwise. If we let

F be the function r in the earlier results of this section, we can find the distribution of $Y = F(X)$. The c.d.f. or Y is, for $0 < y < 1$,

$$G(y) = \Pr(Y \le y) = \Pr(1 - \exp(-X) \le y) = \Pr(X \le -\log(1-y))$$
$$= F(-\log(1-y)) = 1 - \exp(-[-\log(1-y)]) = y,$$

which is the c.d.f. of the uniform distribution on the interval [0, 1]. It follows that Y has the uniform distribution on the interval [0, 1]. ◀

Simulations : Based on ⊛, if we want to simulate a possible value of a random variable with c.d.f. F, you have to:

(a) Generate a random number (uniformly) via a computer, say $Y = y$.

(b) Have a function F of interest and its corresponding F^{-1}.

(c) Then a value little x of some generic X with c.d.f. F is:

$x = F^{-1}(y)$, where you should remember that F^{-1} is the quantile function, and that does not always coincide with the inverse of F.

- Example 3.8.6 page 171 → next page.

Example 3.8.6 Generating Independent Values from a Specified p.d.f. Suppose that a uniform pseudo-random number generator is to be used to generate three independent values from the distribution for which the p.d.f. g is as follows:

$$g(y) = \begin{cases} \frac{1}{2}(2-y) & \text{for } 0 < y < 2, \\ 0 & \text{otherwise.} \end{cases}$$

For $0 < y < 2$, the c.d.f. G of the given distribution is

$$G(y) = y - \frac{y^2}{4}.$$

Also, for $0 < x < 1$, the inverse function $y = G^{-1}(x)$ can be found by solving the equation $x = G(y)$ for y. The result is

$$y = G^{-1}(x) = 2[1 - (1-x)^{1/2}]. \tag{3.8.2}$$

The next step is to generate three uniform pseudo-random numbers x_1, x_2, and x_3 using the generator. Suppose that the three generated values are

$$x_1 = 0.4125, \quad x_2 = 0.0894, \quad x_3 = 0.8302.$$

When these values of x_1, x_2, and x_3 are substituted successively into Eq. (3.8.2), the values of y that are obtained are $y_1 = 0.47$, $y_2 = 0.09$, and $y_3 = 1.18$. These are then treated as the observed values of three independent random variables with the distribution for which the p.d.f. is g. ◀

Functions of 2 or more random variables

Discrete distributions

Reading assigned: Examples 3.9.1, 3.9.2 and Th. 3.9.1.
Not required for the exam.

- Question: Assume that we are given n independent r.v.'s: $X_1, X_2, \ldots, X_n$ and, moreover, we know that each of them has the same distribution, Bernoulli(p).

That is: $X_1 \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}$, $X_2 \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}, \ldots, X_n \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}$.

Let $Y = X_1 + X_2 + \ldots + X_n$.

What is the distribution of the discrete random variable Y?

Answer If $X_1, \ldots, X_n$ i.i.d Bernoulli(p), then $Y = X_1 + \ldots + X_n$ is Binomial(n, p).

Proof See proof of Theorem 3.9.2.

Random variables with continuous joint distributions.

- We focus on bivariate distributions. All results generalize to multivariate distributions.

Theorem 2 Let $(X_1, X_2) \sim f(x_1, x_2)$ continuous.

- Let $Y = h(X_1, X_2)$, for $h: \mathbb{R}^2 \longrightarrow \mathbb{R}$.
- Let $A_y \equiv \{(x_1, x_2) \in \mathbb{R}^2 : h(x_1, x_2) \leq y\}$.

Then: The c.d.f. $G(y)$ of Y is:

$$G(y) = \mathbb{P}(Y \leq y) = \iint_{A_y} f(x_1, x_2)\, dx_1\, dx_2.$$

If G is differentiable, $g(y) = G'(y)$, where $g(y)$ is the p.d.f. of Y.

Note 1. In many important situations we can give a formula for calculating directly the p.d.f. of Y, by generalizing Theorem 1, page 3' above. We will not cover this here (we may return to it!), but you can read the result (not required for exam), Theorem 3.9.5.

Note 2 It is relatively easy to use Theorem 2 directly for some simple transformations, see below.

Theorem 3 If $(X_1, X_2) \sim f(x_1, x_2)$ and $Y = a_1 X_1 + a_2 X_2 + b$, and $a_1 \neq 0$, then $Y \sim g(y)$, with:

$$g(y) = \int_{-\infty}^{\infty} f\left(\frac{y - b - a_2 x_2}{a_1}, x_2\right) \cdot \frac{1}{|a_1|} \, dx_2 .$$

The convolution formula : the distribution of the sum of two independent random variables

Definition Let X_1 and X_2 be independent, continuous r.v.'s. Let $Y = X_1 + X_2$. Then $g(y)$, the distribution of Y, is called the convolution of the p.d.f's of X_1 and X_2.

$$g(y) = \int_{-\infty}^{\infty} f_1(y-z) f_2(z) dz$$

$$= \int_{-\infty}^{\infty} f_1(z) \cdot f_2(y-z) dz .$$

- Example : Example 3.9.5.

Example 3.9.5

An Investment Portfolio. Suppose that an investor wants to purchase both stocks and bonds. Let X_1 be the value of the stocks at the end of one year, and let X_2 be the value of the bonds at the end of one year. Suppose that X_1 and X_2 are independent. Let X_1 have the uniform distribution on the interval [1000, 4000], and let X_2 have the uniform distribution on the interval [800, 1200]. The sum $Y = X_1 + X_2$ is the value at the end of the year of the portfolio consisting of both the stocks and the bonds. We shall find the p.d.f. of Y. The function $f_1(z)f_2(y-z)$ in Eq. (3.9.6) is

$$f_1(z)f_2(y-z) = \begin{cases} 8.333 \times 10^{-7} & \text{for } 1000 \le z \le 4000 \\ & \quad \text{and } 800 \le y - z \le 1200, \\ 0 & \text{otherwise.} \end{cases} \tag{3.9.7}$$

We need to integrate the function in Eq. (3.9.7) over z for each value of y to get the marginal p.d.f. of Y. It is helpful to look at a graph of the set of (y, z) pairs for which the function in Eq. (3.9.7) is positive. Figure 3.25 shows the region shaded. For $1800 < y \le 2200$, we must integrate z from 1000 to $y - 800$. For $2200 < y \le 4800$, we must integrate z from $y - 1200$ to $y - 800$. For $4800 < y < 5200$, we must integrate z from $y - 1200$ to 4000. Since the function in Eq. (3.9.7) is constant when it is positive, the integral equals the constant times the length of the interval of z values. So, the p.d.f. of Y is

$$g(y) = \begin{cases} 8.333 \times 10^{-7}(y - 1800) & \text{for } 1800 < y \le 2200, \\ 3.333 \times 10^{-4} & \text{for } 2200 < y \le 4800, \\ 8.333 \times 10^{-7}(5200 - y) & \text{for } 4800 < y < 5200, \\ 0 & \text{otherwise.} \end{cases}$$

◀

Minimum and maximum of a random sample

- Let $X_1, X_2, \ldots, X_n$ be i.i.d with p.d.f. $f(x)$ and c.d.f. $F(x)$.
- Let $Y_n \equiv \max\{X_1, \ldots, X_n\}$

  $Y_1 \equiv \min\{X_1, \ldots, X_n\}$.
- The respective distributions of Y_1 and Y_n are NOT equal to f (or F).
- We determine them below.
- Let $\begin{cases} G_n \text{ be the c.d.f. of } Y_n \\ g_n \text{ be the pdf of } Y_n \end{cases}$
- Let $\begin{cases} G_1 \text{ be the c.d.f. of } Y_1 \\ g_1 \text{ be the pdf of } Y_1. \end{cases}$

$$G_n(y) = \mathbb{P}(Y_n \le y)$$

$$= \mathbb{P}(X_1 \le y, X_2 \le y, \ldots, X_n \le y)$$

Equality of events

$$= \mathbb{P}(X_1 \le y) \cdot \mathbb{P}(X_2 \le y) \cdot \ldots \cdot \mathbb{P}(X_n \le y)$$

Independence

$$= F(y) \cdot F(y) \cdot \ldots \cdot F(y) = F^n(y)$$

Identically distributed

- Thus, $\boxed{G_n(y) = F^n(y)}$, (*)

$$\boxed{g_n(y) = n \cdot F^{n-1}(y) \cdot f(y)},$$

for all $-\infty < y < \infty$.

Similarly: $\boxed{G_1(y) = 1 - [1 - F(y)]^n}$ and $\boxed{g_1(y) = n[1 - F(y)]^{n-1} \cdot f(y)}$ (**)